Investigation of the Blackbody spectrum

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A system of a spectrometer receiving light input from a light bulb illuminated by a power supply will be utilized in order to investigate three properties of blackbody radiation. Investigation of Wien's displacement law resulted in $(4.896)10E05\ nm/K$, which does not agree with the accepted value of $(2.897)10E06$ nm/K. Investigation of the Stefan-Boltzmann law resulted in (4.11 ± 0.04), which does not agree compared to the accepted value of 4. Investigation of Planck's constant resulted as $(6.64 \pm 0.01)10E-34\ m^2kg/sec$. The accepted value for Planck's constant is $(6.63)10E-34\ m^2kg/sec$, which agrees with the discovered value.

I. BACKGROUND

Blackbody radiation is a phenomenon of an emitted energy from an object at a nonzero temperature via electromagnetic radiation from inside or around the body[1]. Blackbody radiation has three known properties discovered from experimental investigations in the 20th century of Wien's displacement law, Stefan-Boltzmann law, and Planck's constant. Wien's displacement law states that the corresponding wavelength to the maximum frequency will be proportional to the inverse temperature[2]. The Stefan-Boltzmann law states that the total power from the blackbody will be proportional to the fourth power of its temperature[1]. Lastly, spectral intensity is said to follow Planck's law which states the spectra are dependent on a fixed wavelength and temperature. Each of these properties of blackbody radiation will be experimentally investigated.

II. PROPERTY OF WIEN'S DISPLACEMENT LAW

For each measurement set, the software OceanView was used. For collecting current and voltage measurements, an amp-meter will be connected in series and a voltmeter in parallel to the circuit. A spectrometer had a chosen angle of 90 degrees towards the light bulb used to emit light in order to acquire the largest signal. A curtain and cover for the system will be utilized in order to minimize any outside light noise from the spectrometer. To take measurements the voltage will be increased at 15 V intervals. The maximum intensity will be recorded from the software. It is notable that false peaks are observed during measurements. False peaks of intensity are seen in the software at constant wavelengths for all voltages. Measurements are clicked by eye on the program which is a cause for significant uncertainty when the true peaks are unable to be seen due to the false peaks over-

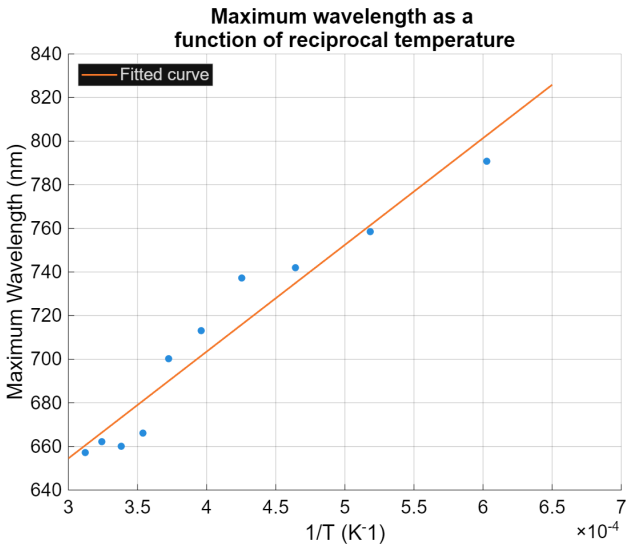


FIG. 1. Wavelength of maximum intensity as a function of reciprocal temperature with a fit line. the fitting function is $\lambda = (4.896)10E05 * T^{-1} + 507.7$

lapping at the same wavelength where the true peak intensity would be. Temperature can be calculated with the relation

$$T(K) = \frac{R_t}{R_r} * (T_r - 60K) + 60K \qquad (1)$$

With T_r being a given room temperature, resistance of the light bulb R_t is calculated with Ohm's Law, and the light bulb's resistance at room temperature is a given value of $(167.93 \pm 0.05)\Omega$. With the calculated temperature, Wien's law will be able to be found by plotting the corresponding wavelength to the maximum frequency at the given voltage λ_{max} as a function of the reciprocal temperature T^{-1}. [2] The fit will be done with the pre-existing MATLAB linear fit function.

A. Resulting Wien's displacement Law

The slope of the resulting fit was $(4.896)10E05\ nm/K$. The accepted value from NIST is $(2.897)10E06$ nm/K[3]. Therefore the calculated value from the data did not agree with the accepted value. Additionally, the χ_r^2 value is 164.26. This indicates the fit is not good for the data. There are several potential sources of error. It is possible light noise from the room may have affected the spectrometer causing erroneous data. Furthermore, it is likely that the data was affected by the false peaks, which caused some of the true measured peaks to be invisible and unfortunately needing to be estimated due to the overlay of the false peaks. Therefore, the true peaks were unable to be precisely or accurately measured. The errors may be resolved by choosing voltages that have peak intensities that are at different wavelengths from the false peaks in future experimentation.

III. PROPERTY OF STEFAN-BOLTZMANN LAW

The same data set of voltage, current, and temperature from the investigation of Wien's Law will be used. An uncertainty estimate for the measurements of current and voltage will be 0.1 V and 0.001 A respectively due to equipment precision limitations. A relationship between varied temperature of the light bulb filament with dissipated power will be investigated

by taking a linear fit of a log-log plot of temperature as a function of power.[1] Power is given by $P = \epsilon A \sigma T^4$, with the Stefan-Boltzmann constant given as $\sigma = 5.67 * 10^{-8}\ W/m^2K^2$. In this experiment, we will measure both voltage and current, which can be used to find the resistance $R = \frac{V}{I}$, and further power with $P = \frac{V^2}{R}$. The resistance has a propagated uncertainty equation of

$$\sigma_R = \sqrt{(\frac{\sigma_v}{I})^2 + (\frac{\sigma_I V}{I^2})^2} \tag{2}$$

Therefore, the power has a calculated uncertainty of

$$\sigma_P = \sqrt{((\frac{2V}{R})\sigma_V)^2 + ((\frac{V^2}{I^2})\sigma_R)^2} \tag{3}$$

A linear fit will be done with a least-squares fit. The resulting slope will reveal the relationship between temperature and power.

A. Resulting Stefan-Boltzmann Law

The resulting slope is (4.11 ± 0.04). According to the Stefan-Boltzmann Law,[1] total power is described as going to the fourth power of temperature. From the results, the power law does not agree within one sigma of the experiment, rather it is 2.75 sigma away. The χ_r^2 value for this fit is 0.17. This could indicate the fit is too good to be true for the data, or that the errors were overestimated. There are multiple potential sources of error within the data. Firstly, the temperature of the bulb itself may not have been perfectly sound due to the system not being perfectly closed. The room temperature potentially affected the true temperature of the bulb for a given voltage due to thermodynamic convection. It may also be possible that the bulb required additional time to settle into it's temperature after receiving energy from the power supply, as measurements were taken as heat was continually increasing instead of cooling.

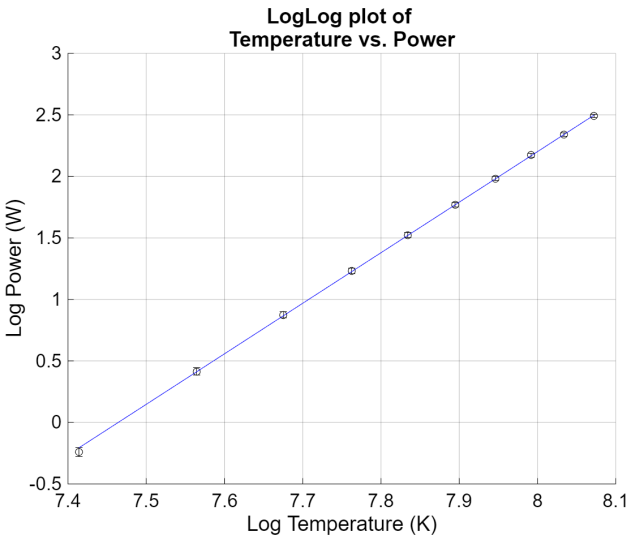


FIG. 2. Log-Log plot of temperature as a function of total power with errorbars and a least squares fitting. The fitting function is $ln(P(K)) = 4.11 * (ln(T(K)) - 30.67$

IV. PROPERTY OF PLANCK'S CONSTANT

The last property to be investigated is Planck's constant. It is expected that spectral intensity will follow Planck's Law for a fixed wavelength and temperature.[4] Constants $c = (3)10E8\ m/s$ for the speed of light and $k = (1.38)10E{-}23\ m^2kg/s^2K$ for the Boltz-

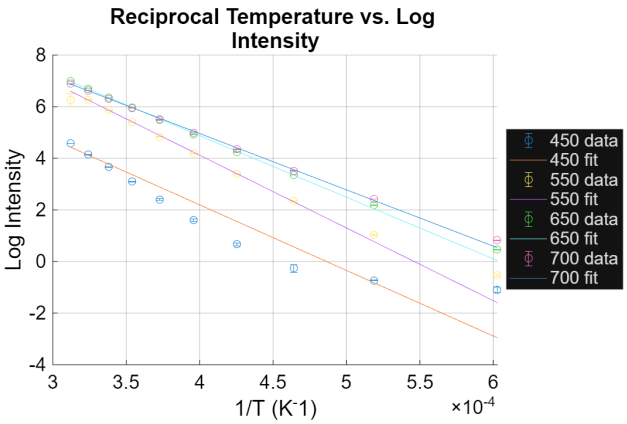


FIG. 3. Reciprocal temperature as a function of log intensity for four fixed wavelengths with error bars and least squares fitting lines. The fitting function is in the form of $ln(I_{meas}) = \frac{m}{T} + B$

mann constant are assumed. The formula

$$ln(I_{meas}) = f(\lambda) - \frac{hc}{\lambda KT} \tag{4}$$

can be used to equate the measured intensity for a fixed wavelength λ and temperature T. The measured intensities are repeatedly measured to extract uncertainty from the standard deviation. The natural log of measured intensity can be plotted against T^{-1} to extract Planck's constant h. Fitting methods and will follow the above procedures using least squares fitting with intention to assign the Y-axis to the variable with dominant error. The uncertainty is seen to increase as the voltage decreases in the experiment.

A. Results of Planck's Constant

For each measured wavelength a single Planck's constant value was extracted. They will be finally averaged to conclude on a single constant result.

Wavelength (nm)	Planck's constant m^2kg/sec	χ_r^2
450	$(5.2656 \pm 0.0016)10E-34$	190.29
550	$(7.1375 \pm 0.0011)10E-34$	135.16
650	$(7.1399 \pm 0.0004)10E-34$	157.38
700	$(7.0321 \pm 0.0004)10E-34$	262.53

TABLE I. Wavelength, Planck's constant with uncertainty, and reduced chi squared values resulting from least squares fittings.

After fitting and extracting each Planck constant from each fixed wavelength at a certain temperature, the result of the averaged Planck constant is $(6.64 \pm 0.01)10E-34\ m^2kg/sec$. The accepted value for Planck's constant from NIST is $(6.63)10E-34\ m^2kg/sec$,[5] which would agree with the calculated value. However, each χ_r^2 value for each fit is very high. This would indicate that the fit is not good for the data or that the errors were underestimated. It is likely that systemic errors still would have occurred, perhaps with imperfect shielding methods of the optic sensor from sources of light outside of the system. It is also possible that previously mentioned sources of error would apply to this data. It can be concluded that despite the agreed result, the χ_r^2 may indicate that the fit would not be good for the data used.

V. CONCLUSION

With the investigation of the Wien Displacement Law, it was found that the constant

from taking the corresponding wavelength to the maximum intensity as a function of T^{-1} was $(4.896)10E05\ nm/K$, while the accepted value from NIST is $(2.897e+06)\ nm/K$. The χ_r^2 value was 164.26. This indicates the fit is not good for the data. The false-peak signals from the spectrometer are the most likely cause of the errors, since the true peak wavelength had to be estimated by eye in the program. Other sources of error from the room such as light noise may have also affected the results. By investigating the Stefan-Boltzmann law, it was found that the slope of the fitted Log-Log plot, corresponding to the exponential relationship of power to temperature, was (4.11 ± 0.04). This is within three sigma of the known value of 4. The χ_r^2 value was 0.17. This could indicate the fit is too good to be true for the data, or there was overestimation within the errors. Systemic errors have likely contributed to the slight error of the result, such as interactions of temperatures outside of the system affecting the temperature of the bulb, or perhaps the temperature was not given adequate time to settle. Lastly, when investigating Planck's constant it was found that $(6.64 \pm 0.01)10E-34\ m^2kg/sec$ was obtained as Planck's constant from the average of each calculated value at fixed wavelengths, and the accepted value for Planck's constant from NIST is $(6.63)10E-34\ m^2kg/sec$, which would agree with the findings. The resulting χ_r^2 values for each fit were all much greater than one, which may indicate the errors were under-estimated or the fit is not good for the data. Some sources of error may have come from light noise from an imperfectly closed system, or same as previously mentioned.

[1] Wellons, M. (2007). The stefan-boltzmann law. Physics Department, The College of Wooster, Wooster, Ohio, 44691, 25. https://physics.wooster.edu/wp-content/uploads/2021/08/Junior-IS-Thesis-Web_2007_Wellons.pdf, Retrieved April 13, 2026.

[2] Stewart, S. M. (2012). Spectral peaks and Wien's displacement law. Journal of Thermophysics and Heat Transfer, 26(4), 689-692. https://arc.aiaa.org/doi/pdf/10.2514/1.T3789, Retrieved April 13, 2026.

[3] NIST website with fundamental constants of Wien's Displacement Law, https://physics.nist.gov/cgi-bin/cuu/Value?bwien, Retrieved April 10, 2026.

[4] Steiner, R. (2013). History and progress on accurate measurements of the Planck constant. Reports on Progress in Physics, 76(1), 016101. https://iopscience.iop.org/article/10.1088/0034-4885/76/1/016101/pdf, Retrieved April 13, 2026.

[5] NIST website with fundamental constants of Planck's constant, https://www.nist.gov/physics/explainers/planck-constant, Retrieved April 10, 2026.